

- 1 Algae are a diverse group of photosynthetic eukaryotes in the kingdom Protocista.

Rhodomonas salina and *Skeletonema costatum* are two species of alga that live in seawater.

Fig. 1.1 shows a scanning electron micrograph of *R. salina*, which is a unicellular organism. *R. salina* is red in colour.

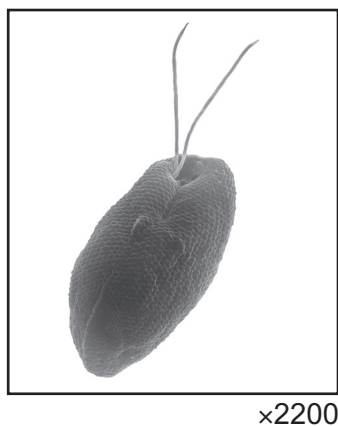


Fig. 1.1

Fig. 1.2 shows a scanning electron micrograph of *S. costatum*, which is also a unicellular organism. The individual cells can group together in a chain. *S. costatum* is yellow-brown in colour.

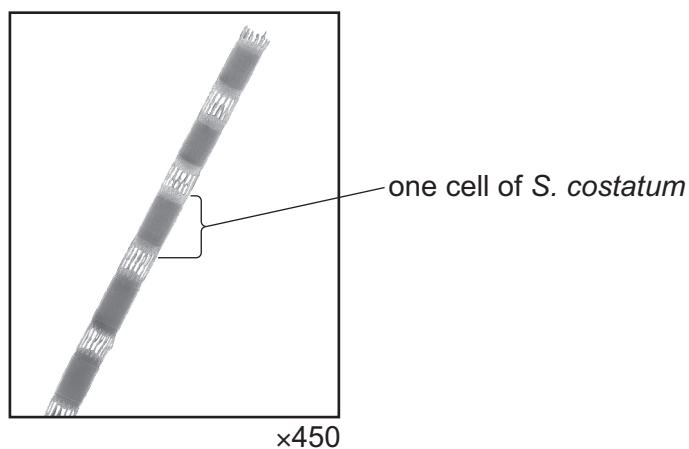


Fig. 1.2

- (a) A student prepared extracts of the photosynthetic pigments from *R. salina* and *S. costatum*. The student carried out chromatography to identify and compare the pigments in the two species.

The chromatograms are shown in Fig. 1.3.

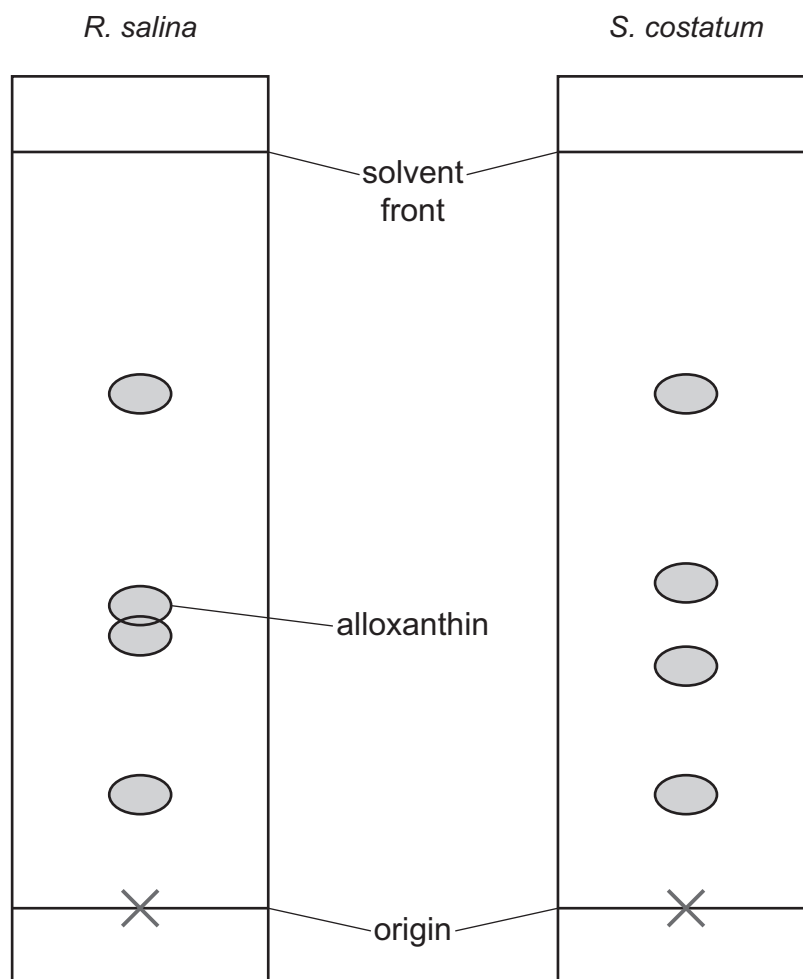


Fig. 1.3

- (i) Use Fig. 1.3 to calculate the R_f value of alloxanthin from *R. salina*.

R_f of alloxanthin = [1]

(ii) Table 1.1 shows the R_f values of some photosynthetic pigments.

Table 1.1

photosynthetic pigment	R_f value
β -carotene	0.94
chlorophyll a	0.68
chlorophyll b	0.54
chlorophyll c	0.15
diadinoxanthin	0.32
fucoxanthin	0.43
phycocyanin	0.35

R. salina lacks two photosynthetic pigments that are present in *S. costatum*.

Use Fig. 1.3 and Table 1.1 to identify the **two** photosynthetic pigments that are present in **only** *S. costatum*.

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..... [2]

(b) The extract of the photosynthetic pigments from each of the two species was used to obtain absorption spectra as shown in Fig. 1.4.

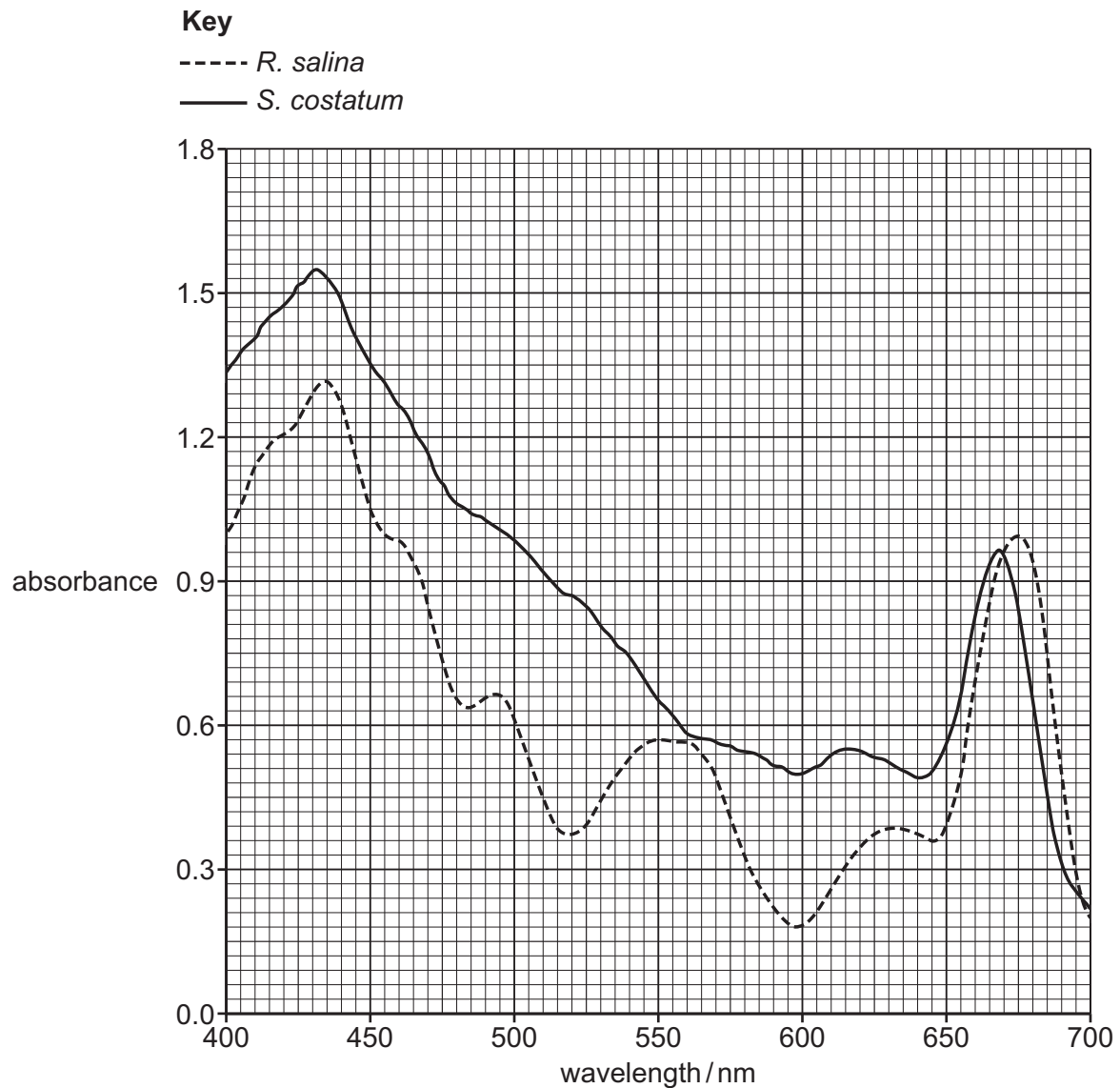


Fig. 1.4

The student planned to investigate the effect of light wavelength on the rate of photosynthesis in *R. salina* and *S. costatum*.

The student planned to determine the rate of photosynthesis in the two species when they were exposed to three different colours of light:

- blue light, peak wavelength of 455 nm
- green light, peak wavelength of 550 nm
- red light, peak wavelength of 670 nm.



Use Fig. 1.4 to compare the expected rates of photosynthesis for the two species in blue, green and red light.

blue light

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green light

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red light

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[3]



- (c) In the investigation, the student used three different filters, blue, green and red, to expose the two species to the three different colours of light.

The student immobilised the algae in sodium alginate to form two sets of algal beads, one set for *R. salina* and one set for *S. costatum*.

The algal beads were the same size. Photosynthesis of the algae is **not** affected when they are immobilised in the beads.

The student placed the algal beads in small bottles for the experiment as shown in Fig. 1.5.

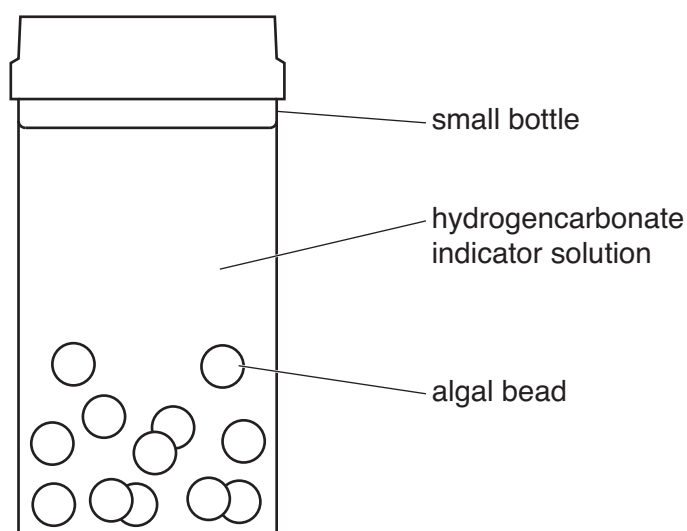


Fig. 1.5

The student used hydrogencarbonate indicator solution to estimate the rate of photosynthesis when the algal beads were exposed to each colour of light. The indicator solution changes colour with pH as shown in Table 1.2.

Table 1.2

colour of indicator solution	pH	CO ₂ concentration	rate of photosynthesis
yellow	7.6	↑ increasing concentration	↑ increasing rate
yellow-orange	7.8		
orange	8.0		
orange-red	8.2		
red	8.4	atmospheric concentration (0.04%)	
red-magenta	8.6	↓ decreasing concentration	
magenta	8.8		
magenta-purple	9.0		
purple	9.2		

The student used pH change as an indirect measure of the rate of photosynthesis. A greater decrease in CO₂ concentration indicates a higher rate of photosynthesis.

For each bottle tested, the student:

- added algal beads to a small bottle containing indicator solution
- removed a sample of the indicator solution from the small bottle after some time
- used a colorimeter to measure the absorbance of the sample of the indicator solution.

The student used a calibration curve to estimate the pH of the samples of the indicator solution.

- (i) Suggest **one** reason why using algal beads, instead of placing the algal cells directly into the indicator solution in the bottles, improves the validity of the investigation.

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..... [1]

- (ii) Identify the **two independent** variables and the **dependent** variable in this investigation.

independent variable 1

independent variable 2

dependent variable

[3]

[illegible]

The student carried out the investigation in a temperature-controlled room.

Describe a method the student could use to collect sufficient data so that the rates of photosynthesis of both species of alga in the three colours of light can be compared.

The description of your method should be set out in a logical way and be detailed enough for another person to follow.

You should **not** include a risk assessment or details of how to prepare the algal beads.

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

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- (iv) Identify a hazard in this investigation **and** state a risk associated with the hazard **and** state one precaution that the student should take.

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- (d) For each colour of light, the student carried out a statistical test to compare the pH of the samples taken from the small bottles that contained *R. salina* and *S. costatum*.

The student decided that a *t*-test was the most appropriate test to use for these data.

- (i) Suggest **one** reason why a *t*-test was the most appropriate test to use for these data.

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..... [1]

- (ii) State a null hypothesis for this test.

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[Total: 21]